DELHI PUBLIC SCHOOL NEWTOWN
SESSION 2023–24
PRE BOARD EXAMINATION

CLASS: X | **FULL MARKS: 80**

SUBJECT: MATHEMATICS [SET A] | **TIME: $2\frac{1}{2}$ HOURS**

Answers to this Paper must be written on a paper provided separately.

You will not be allowed to write during the first 15minutes.

This time is to be spent in reading the question paper.

The time given at the head of this Paper is the time allowed for writing the answers.

Attempt all questions from Section A and any four questions from Section B.

***The intended marks for questions or parts of questions are given in brackets* [ ].**

This paper contains 8 printed pages.

SECTION A

(*Attempt* ***all*** *questions from this section)*

Question 1

Choose the correct answers to the questions from the given options. (Do not copy the question, Write the correct answer only.) **[15]**

(i) The tax invoice of a telecom service in Ropar shows cost of services provided by it as ₹ 750. The total bill amount including GST is ₹ 885, the rate of CGST is:

(a) 5% **(b) 9%** **(c) 14%** **(d) 18%**

(ii) Which of the following is a quadratic equation?
(a) $x^2 + 2x + 1 = (4 - x)^2 + 3$
(b) $-2x^2 = (5 - x)(2x - 25)$
(c) $(k + 1)x^2 + 32x = 7$, where $k = -1$
(d) $x^3 - x^2 = (x - 1)^3$

(iii) If $A = \begin{bmatrix} 3 & 1 \\ 0 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & 1 \\ 0 & 2 \end{bmatrix}$, then which of the following is/are correct?

I. AB is defined **II. BA is defined** **III. AB = BA**

Select the correct answer

(a) I only **(b) II only** **(c) Both I and II** **(d) I, II and III**

(iv) Given that $x^2 - x - 6$ is a factor of $x^3 - 7x - 6$, then the remaining factor is:
(a) (x + 1) (b) x (c) (x - 2) (d) (2x + 1)

(v) If S_r denotes the sum of first r terms of an A.P. then S_{3n} : (S_{2n} - S_n) is:

(a) 3n:5 (b) 3n:1 (c) 3n :2 (d)3 :1

(vi) If the points P(1, 2), B(0, 0) and C(a, b) are collinear, then
(a) 2a = b (b) a = -b (c) a = 2b (d) a = b

(vii) Diagonals of a trapezium PQRS intersect each other at the point O, PQ || RS and PQ = 3 RS, Then the ratio of areas of triangles POQ and ROS is:

(a) 1:9 (b) 9:1 (c) 3:1 (d) 1:3

(viii) The radius of a piece of wire is decreased to one-half. If the volume remains the same, then its length will increase:

(a) 2 times (b) 3 times (c) 4 times (d) 5 times

(ix) If p, q, r, s are in proportion then:
(a) (p + q): q = (r - s): s (b) (p + q): q = (r + s): s
(c) (p + q): s = (r - s): q (d) (p + q): q = (p - q): q

(x) Event A: It will rain in Mumbai in the month of July.

Event B: Thursday will be the day after Friday next week.

Event C: There are 30 days in the month of September.

Which of the above event(s) has probability equal to 1?

(a) all events A, B and C (b) both events A and B

(c) both events B and C (d) both events A and C

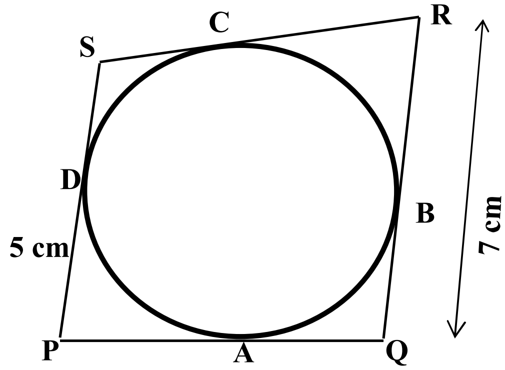
(xi) If 6 is the root of the equation $x^2 + bx + 12 = 0$ and the equation $x^2 + bx + q = 0$ has equal roots, then the value of q is:

(a) 8 (b) -8 (c) 16 (d) -16

(xii) If ΔABC is right angled at C, then the value of cos(A+B) is :

(a) 0 (b) 1 (c) ½ (d) √3/2

(xiii) In the given figure, the sides of the quadrilateral PQRS touches the circle at A, B, C and D. If RC = 4 cm, RQ = 7 cm and PD = 5 cm, the length of PQ is:



(a) 5cm (b) 6cm (c) 7cm (d) 8cm

(xiv) 1. Shares of a company A, paying 15%, ₹ 100 shares are at ₹ 60.
2. Shares of a company B, paying 15%, ₹ 100 shares are at ₹ 100.
3. Shares of a company C, paying 15%, ₹ 100 shares are at ₹ 130.
Shares of which company are at discount?
(a) Company A (b) Company B (c) Company C (d) Company A and C

(xv) Assertion (A) : Frequency is the number of times a particular observation occurs in data.
Reason (R) : Data can be grouped into class intervals such that all observations in that range belong to that class.

(a) A is true, R is false
(b) A is false, R is true
(c) both A and R are true
(d) both A and R are false

Question 2

(i) A storage tank consists of a circular cylinder, with a hemisphere adjoined on either end. If the external diameter of the cylinder be 1.4m and its length be 5m, what will be the cost of painting it on the outside at the rate of ₹10 per square metre? [4]

(ii) In a recurring deposit account for 2 years, the total amount deposited by Mr Patel is ₹ 18,000. If the interest earned by him is one–sixth of his total deposit, then find:
a) the interest he earns. b) his monthly deposit. c) the rate of interest. [4]

(iii) Find:
a) $(\sin\theta + \sec\theta)^2$ b) $(\cos\theta + \mathrm{cosec}\,\theta)^2$
Using the above results prove the following trigonometry identity.
$(\sin\theta + \sec\theta)^2 + (\cos\theta + \mathrm{cosec}\,\theta)^2 = (1 + \sec\theta\,cosec\theta)^2$ [4]

Question 3

(i) If x : a = y : b , show that $\frac{x^2+a^2}{x+a} + \frac{y^2+b^2}{y+b} = \frac{(x+y)^2+(a+b)^2}{(x+y)+(a+b)}$ **[4]**

(ii) In the given figure, O is the centre of the circle. PQ is a tangent to the circle at T. Chord AB produced meets the tangent at P. AB = 9cm, BP = 16cm, <PTB = 50º <OBA = 45º.

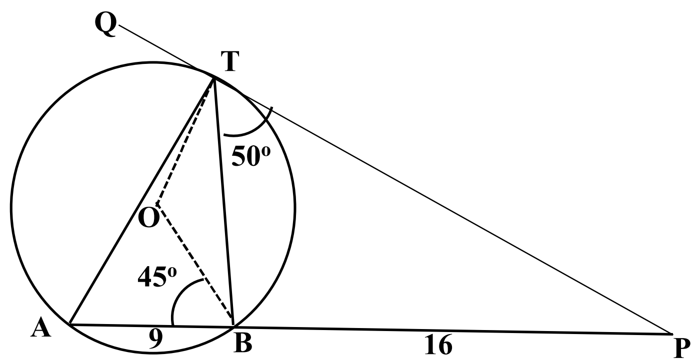
Find:

(a) Length of PT

(b) ∠BAT

(c) ∠BOT

(d) ∠ABT



[4]

(iii) Use graph paper to answer this question.
During a medical checkup of 60 students in a school, weights were recorded as follows:

Weight (in kg)	28 - 30	30 - 32	32 - 34	34 - 36	36 - 38	38 - 40	40 - 42	42 - 44
No: of students	2	4	10	13	15	9	5	2

Taking 2cm = 2kg along one axis and 2cm = 10 students along the other axis draw an ogive. Use your graph to find the
Hence use your graph to find the following:
a) Median
b) Upper Quartile
c) Number of students whose weight is above 37 kg. **[5]**

SECTION B

(*Attempt **any four** questions from this section)*

Question 4

(i) Solve the quadratic equation $2x^2 - 1 = 7x$ for x and give your answer correct to two places of decimal. **[3]**

(ii)If A = $\begin{bmatrix} sin\theta & cos\theta \\ cos\theta & -sin\theta \end{bmatrix}$, show that $A^2 = I$, where I is the unit matrix and θ =90^0.

[3]

(iii) ABCD is a parallelogram. P is a point on BC such that BP: PC = 1: 2 and DP produced meets AB produced at Q. If the area of ($\Delta\, CPQ$) = 20 cm^2, find:

a) area of ($\Delta\, BPQ$)

b) area of ($\Delta\, CPD$)

c) area of (parallelogram ABCD)

[4]

Question 5

(i) Use the step deviation method to find the mean household expenditure (in rupees) of the following distribution.

Expenditure (in ₹)	100-150	150-200	200-250	250-300	300-350	350-400	400-450	450-500
Frequency (f)	15	50	80	76	72	45	39	9

[3]

(ii) The following bill shows the GST rate and the marked price of articles:

Chawla Electronics				
S. No:	Item	Marked Price	Discount	Rate of GST
(a)	Refrigerator (256 litres)	₹ 30,000	15%	18%
(b)	Microwave Oven	₹ 25,600	Nil	28%

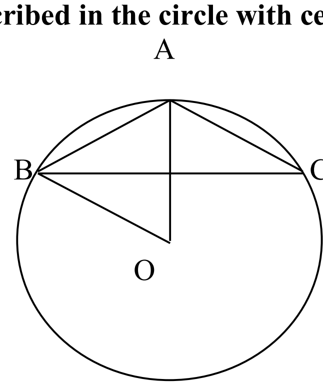
Find the total amount to be paid (including GST) for the above bill. [3]

(iii) In the figure given, AB is a side of a regular six sided polygon and AC is a side of a regular eight sided polygon inscribed in the circle with centre O. Find giving suitable reasons, the measure of:

(a) ∠ AOB

(b) ∠ACB

(c) ∠ ABC



[4]

Question 6

(i) The sum of the first three terms of a G.P. is 16 and the sum of the next three terms is 128. Determine the first term and the common ratio of the G.P.

[3]

(ii) Solve the following equation for x , using the properties of proportion:

$$\frac{x^4+1}{2x^2} = \frac{17}{8}$$

[3]

(iii) A solid is in shape of a hemisphere of radius 7cm, surmounted by a cone of height 4 cm. The solid is immersed completely in a cylindrical container filled with water to a certain height. If the radius of the cylinder is 14cm, find the rise in water level.

[4]

Question 7

(i) If $x^3 + ax^2 - x + b$ has (x – 2) as a factor and leaves a remainder 3 when divided by (x – 3), find a and b. **[3]**

(ii) A die is thrown 200 times and the outcomes 1, 2, 3, 4, 5, 6 have frequencies as below:

Outcome	1	2	3	4	5	6
Frequency	40	38	43	29	28	22

Find the probability of
a) getting a prime number b) getting an odd number c) getting a multiple of 3.

[3]

(iii) Use a graph sheet for the question, take 2cm = 1 unit along both x and y axis:
a) Plot the points A (3,2) and B (5, 0). Reflect point A on the y- axis to A'. Write the co-ordinates of A'.
b) Reflect point B on the line AA' to B'. Write the coordinates of B'.
c) Give a geometrical name to the closed figure A'B'ABA'. **[4]**

Question 8

(i) Solve the following inequation and represent the solution set on the number line.

$$-\frac{x}{3} - 4 < \frac{x}{2} - \frac{7}{3} \leq \frac{7}{6},\ x \in W$$

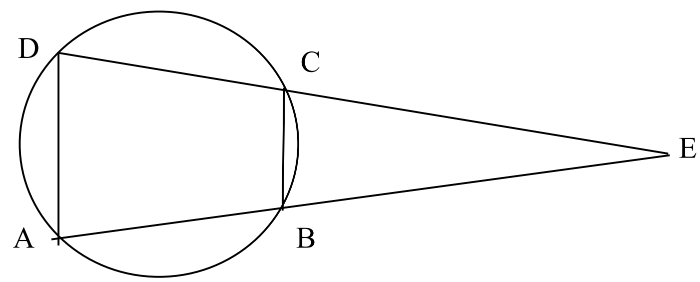
[3]

(ii) In the given figure ABCD is a cyclic quadrilateral. AB and DC are produced to meet at E.

(a) Prove Δ EBC ~Δ EDA

(b) Find AB if CD = 2cm, CE = 6cm and BE = 3cm

(c) Find $\dfrac{\textbf{Area (Quad ABCD)}}{\textbf{Area }(\Delta\, EDA)}$



[3]

(iii) Two trains leave a railway station at the same time. The first train travels due west and the second train due north. The first train travels 5km/hr faster than the second train. If after two hours, they are 50km apart, find the average speed of each train. **[4]**

Question 9

(i) Abhay invests ₹ 22,500 in ₹ 50 shares available at 10% discount. If the dividend paid by the company is 12%, calculate:

a) The number of shares purchased

b) The annual dividend received.

c) The rate of return he gets on his investment. Give your answer correct to the nearest whole number. **[3]**

(ii) A (2, 3), B(4, 5) and C(6, 6) are the coordinates of the vertices of a triangle; D is the midpoint of BC; G is a point on AD such that: AG : GD = 2: 1. Find:

a) the equation of median AD;

b) the coordinates of G. **[3]**

(iii) Construct a triangle ABC in which BC = 6cm, AB = 5.5cm and <ABC = 120^0. Construct a circle circumscribing the triangle ABC. Draw a cyclic quadrilateral ABCD so that D is equidistant from B and C. **[4]**

Question 10

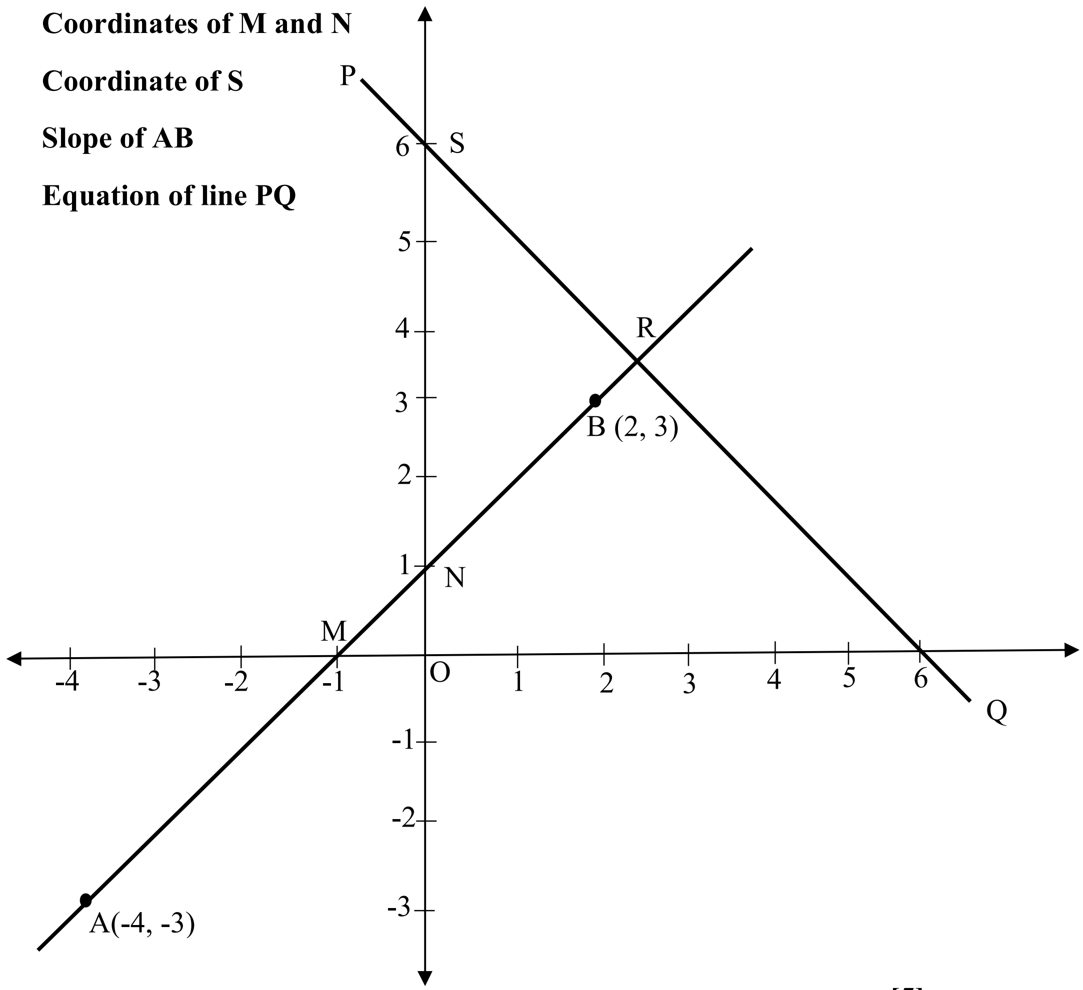
(i) **The line segment joining A (-4, -3) and B (2, 3) is intercepted by the x-axis at the point M and the y-axis at the point N. PQ is perpendicular to AB at R and meets the y-axis at a distance of 6 units from the origin O, as shown in the diagram, at S. Find the:**

(a) **Coordinates of M and N**

(b) **Coordinate of S**

(c) **Slope of AB**

(d) **Equation of line PQ**



[5]

(ii) A 1.2m tall girl spots a balloon moving with the wind in a horizontal line at a height of 88.2m from the ground. The angle of elevation of the balloon from the eyes of the girl at any instant is 60^0. After some time, the angle of elevation reduces to 30^0. Find the distance travelled by the balloon during the interval. **[5]**